Integrale Gamma și Beta

Calculați următoarele integrale, folosind integralele Gamma și Beta.

1. $I = \int_0^{\infty} x^5 e^{-2x} dx.$

Notăm: $2x = t \Rightarrow dx = \frac{1}{2} dt$; $x = \frac{t}{2}$

$x = 0 \Rightarrow t = 0$; $x = +\infty \Rightarrow t = +\infty$

$$\Rightarrow I = \frac{1}{2} \int_0^{\infty} \left(\frac{t}{2}\right)^5 e^{-t} dt = \frac{1}{2^6} \int_0^{\infty} t^5 e^{-t} dt = \frac{1}{2^6} \Gamma(6) =$$

$$= \frac{5!}{2^6} = \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5}{2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2} = \frac{15}{8}$$

2. $I = \int_{-\infty}^{\infty} x^6 e^{-x^2} dx$

Funcția $f(x) = x^6 \cdot e^{-x^2}$ este pară $\Rightarrow \int_{-\infty}^{\infty} f(x) dx =$

$$= 2 \int_0^{\infty} f(x) dx.$$

Notăm: $x^2 = t \Rightarrow 2x dx = dt \Rightarrow 2 t^{\frac{1}{2}} dx = dt \Rightarrow$

$\Rightarrow dx = \frac{1}{2} t^{-\frac{1}{2}} dt$; $x = 0 \Rightarrow t = 0$; $x = +\infty \Rightarrow t = +\infty$ $(x = t^{\frac{1}{2}})$

$$\Rightarrow I = 2 \cdot \int_0^{\infty} t^3 \cdot e^{-t} \cdot \frac{1}{2} \cdot t^{-\frac{1}{2}} dt = \int_0^{\infty} t^{\frac{5}{2}} e^{-t} dt =$$

$$= \Gamma\left(\frac{7}{2}\right) = \Gamma\left(\frac{5}{2}+1\right) = \frac{5}{2} \Gamma\left(\frac{5}{2}\right) = \frac{5}{2} \Gamma\left(\frac{3}{2}+1\right) =$$

$$= \frac{15}{4} \Gamma\left(\frac{3}{2}\right) = \frac{15}{4} \Gamma\left(\frac{1}{2}+1\right) = \frac{15}{8} \Gamma\left(\frac{1}{2}\right) = \frac{15\sqrt{\pi}}{8}$$

3. $I = \int_{-1}^{\infty} \sqrt{x+1}\, e^{-x-1} dx.$

Notăm: $x+1 = t \Rightarrow dx = dt$; $x = -1 \Rightarrow t = 0$; $x = +\infty \Rightarrow$

$\Rightarrow t = +\infty$

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$$\Rightarrow I = \int_0^{\infty} t^{\frac{1}{2}} e^{-t} dt = \Gamma\left(\frac{3}{2}\right) = \Gamma\left(\frac{1}{2}+1\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}$$

4. $I = \int_0^1 \sqrt{x} \ln^3 x \, dx$

Notăm: $\ln x = t \Rightarrow \frac{1}{x}dx = dt ; x = e^t \Rightarrow dx = e^t dt$

$x = 0 \Rightarrow t = -\infty ; x = 1 \Rightarrow t = 0$

$$\Rightarrow I = \int_{-\infty}^{0} e^{\frac{t}{2}} \cdot t^3 \cdot e^t dt = \int_{-\infty}^{0} e^{\frac{3t}{2}} \cdot t^3 dt$$

Notăm: $\frac{3t}{2} = -y \Rightarrow \frac{3}{2}dt = -dy \Rightarrow dt = -\frac{2}{3}dy$

$\Rightarrow t = -\frac{2}{3}y$

$t = -\infty \Rightarrow y = \infty ; t = 0 \Rightarrow y = 0$

$$\Rightarrow I = \int_{\infty}^{0} e^{-y} \cdot \left(-\frac{8}{27}\right) y^3 \cdot \left(-\frac{2}{3}\right) dy =$$

$$= -\frac{16}{81} \int_0^{\infty} y^3 e^{-y} dy = -\frac{16}{81}\Gamma(4) = -\frac{16}{81} \cdot 3! =$$

$$= -\frac{32}{27}.$$

5. $I = \int_1^{\infty} \frac{\ln x}{x^a} dx ; a > 1$

Notăm: $\ln x = t \Rightarrow \frac{1}{x}dx = dt ; x = e^t \Rightarrow dx = e^t dt$

$x = 1 \Rightarrow t = 0 ; x = \infty \Rightarrow t = \infty$

$$\Rightarrow I = \int_0^{\infty} t \cdot e^{-ta} \cdot e^t dt = \int_0^{\infty} t \cdot e^{t(1-a)} dt$$

Notăm: $t(a-1) = y \Rightarrow (a-1)dt = dy \Rightarrow dt = \frac{1}{a-1}dy$

$t = 0 \Rightarrow y = 0 ; t = \infty \Rightarrow y = +\infty$

$$\Rightarrow I = \int_0^{\infty} \frac{y}{a-1} \cdot e^{-y} \cdot \frac{1}{a-1} dy = \frac{1}{(a-1)^2} \int_0^{\infty} y e^{-y} dy =$$

$$= \frac{1}{(a-1)^2}\Gamma(2) = \frac{1}{(a-1)^2}.$$

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6. $I = \int_0^1 \sqrt{x-x^2}\,dx = \int_0^1 \sqrt{x(1-x)}\,dx = \int_0^1 x^{\frac{1}{2}}(1-x)^{\frac{1}{2}}dx =$

$= B\left(\frac{3}{2}, \frac{3}{2}\right) = \frac{\Gamma\left(\frac{3}{2}\right)\cdot\Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{3}{2}+\frac{3}{2}\right)} =$

$= \frac{\left[\Gamma\cdot\left(\frac{1}{2}+1\right)\right]^2}{\Gamma(3)} = \frac{\left[\frac{1}{2}\Gamma\left(\frac{1}{2}\right)\right]^2}{2!} = \frac{\left(\frac{1}{2}\sqrt{\pi}\right)^2}{2} =$

$= \frac{\pi}{8}$

7. $I = \int_0^1 \frac{1}{\sqrt[3]{x^2(1-x)}}\,dx = \int_0^1 x^{-\frac{2}{3}}(1-x)^{-\frac{1}{3}}dx =$

$= B\left(\frac{1}{3}, \frac{2}{3}\right) = \frac{\Gamma\left(\frac{1}{3}\right)\cdot\Gamma\left(\frac{2}{3}\right)}{\Gamma\left(\frac{1}{3}+\frac{2}{3}\right)} = \frac{\Gamma\left(\frac{1}{3}\right)\cdot\Gamma\left(1-\frac{1}{3}\right)}{\Gamma(1)} =$

$= \frac{\pi}{\sin\frac{1}{3}\pi} = \frac{\pi}{\frac{\sqrt{3}}{2}} = \frac{2\pi\sqrt{3}}{3}$

8. $I = \int_{-\infty}^{\infty} e^{2x}\left(1+e^{2x}\right)^{-2}dx$

Notăm: $e^{2x} = t \Rightarrow 2e^{2x}dx = dt \Rightarrow e^{2x}dx = \frac{1}{2}dt \Rightarrow$

$\Rightarrow dx = \frac{1}{2t}dt$; $x = -\infty \Rightarrow t = 0$; $x = +\infty \Rightarrow t = +\infty$

$\Rightarrow I = \int_0^{\infty} t(1+t)^{-2}\cdot\frac{1}{2t}dt = \frac{1}{2}\int_0^{\infty}(1+t)^{-2}dt =$

$= B(1,1) = \frac{\Gamma(1)\cdot\Gamma(1)}{\Gamma(1+1)} = \frac{1}{\Gamma(2)} = \frac{1}{1!} = 1$

9. $I = \int_1^e \frac{1}{x}\ln^3 x\left(1-\ln x\right)^4 dx.$

Notăm: $\ln x = t \Rightarrow \frac{1}{x}dx = dt$

$x = 1 \Rightarrow t = 0$; $x = e \Rightarrow t = 1$

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$$\Rightarrow I = \int_0^1 t^3(1-t)^4\,dt = B(4,5) = \frac{\Gamma(4)\cdot\Gamma(5)}{\Gamma(9)} =$$

$$= \frac{3!\cdot 4!}{8!} = \frac{1\cdot 2\cdot 3\cdot 2\cdot 3\cdot 4}{2\cdot 3\cdot 4\cdot 5\cdot 6\cdot 7\cdot 8} = \frac{1}{280}$$

10. $I = \int_0^{\infty} \frac{x}{1+x^6}\,dx$

Notăm: $x^6 = t \Rightarrow 6x^5dx = dt;\ x = t^{\frac{1}{6}} \Rightarrow dx = \frac{1}{6}t^{-\frac{5}{6}}dt$

$x = 0 \Rightarrow t = 0;\ x = +\infty \Rightarrow t = +\infty$

$$\Rightarrow I = \int_0^{\infty} t^{\frac{1}{6}}\cdot(1+t)^{-1}\cdot\frac{1}{6}t^{-\frac{5}{6}}dt = \frac{1}{6}\int_0^{\infty}\frac{t^{-\frac{2}{3}}}{1+t}\,dt =$$

$$= \frac{1}{6}B\left(\frac{1}{3},\frac{2}{3}\right) = \frac{1}{6}\cdot\frac{\Gamma\left(\frac{1}{3}\right)\Gamma\left(\frac{2}{3}\right)}{\Gamma(1)} = \frac{1}{6}\cdot\frac{\pi}{\sin\frac{\pi}{3}} =$$

$$= \frac{1}{6}\cdot\frac{2\pi\sqrt{3}}{3} = \frac{\pi\sqrt{3}}{9}$$

11. $I = \int_{-4}^{3} \frac{1}{\sqrt[6]{(x+4)(3-x)^5}}\,dx = \int_{-4}^{3}(x+4)^{-\frac{1}{6}}(3-x)^{-\frac{5}{6}}dx$

Notăm: $\frac{x+4}{7} = t \Rightarrow dx = 7dt;\ x = 7t-4$

$x = -4 \Rightarrow t = 0;\ x = 3 \Rightarrow t = 1$

$$\Rightarrow I = \int_0^1 (7t)^{-\frac{1}{6}}\cdot\left[7(1-t)\right]^{-\frac{5}{6}}\cdot 7dt =$$

$$= \int_0^1 t^{-\frac{1}{6}}\cdot(1-t)^{-\frac{5}{6}}dt = B\left(\frac{5}{6},\frac{1}{6}\right) =$$

$$= \frac{\Gamma\left(\frac{5}{6}\right)\cdot\Gamma\left(\frac{1}{6}\right)}{\Gamma(1)} = \frac{\pi}{\sin\frac{\pi}{6}} = 2\pi.$$

12. $I = \int_0^1 \frac{1}{\sqrt[m]{1-x^m}}\,dx$

Notăm: $x^m = t \Rightarrow mx^{m-1}dx = dt \Rightarrow dx = \frac{1}{m}t^{\frac{1}{m}-1}dt$

$x = 0 \Rightarrow t = 0;\ x = 1 \Rightarrow t = 1$

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$$\Rightarrow I = \int_0^1 (1-t)^{-\frac{1}{n}} \cdot \frac{1}{n} \cdot t^{\frac{1}{n}-1} dt = \frac{1}{n} \int_0^1 t^{\frac{1}{n}-1} \cdot (1-t)^{-\frac{1}{n}} dt =$$

$$= \frac{1}{n} B\left(\frac{1}{n}, \frac{n-1}{n}\right) = \frac{1}{n} \frac{\Gamma\left(\frac{1}{n}\right) \cdot \Gamma\left(\frac{n-1}{n}\right)}{\Gamma(1)} =$$

$$= \frac{1}{n} \cdot \frac{\pi}{\sin \frac{\pi}{n}} .$$

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